Climate Sensitivity

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Climate sensitivity

- It is a measure of how much Earth's surface will cool or warm after a specified factor causes a change in its climate system, such as how much it will warm for a doubling in carbon dioxide (CO_2) concentration in the atmospheric .
- In technical terms, climate sensitivity is the average change in global mean surface temperature in response to a radiative forcing, which drives a difference between Earth's incoming and outgoing energy.
- Climate sensitivity is a key measure in climate science, and a focus area for climate scientists, who want to understand the ultimate consequences of anthroprogenic global warming.
- There are several ways of defining climate sensitivity, depending on the timescales of interest. Two of those are:
- **Transient Climate Response (TCR)**: The temperature increase at the instant that atmospheric CO_2 has doubled (following an increase of 1% each year) gives us the Transient Climate Response. This is useful as a gauge for what we might expect over the current century when atmospheric concentrations of CO_2 are changing.

- **Equilibrium Climate Sensitivity (ECS)**: The climate system will continue to warm for some time after the TCR point, largely as the oceans are very slow to respond.

- Therefore we can also consider the temperature increase that would eventually occur (after hundreds or even thousands of years) when the climate system fully adjusts to a sustained doubling of CO_2 - this is called the Equilibrium Climate Sensitivity.

- The long timescales involved here mean ECS is arguably a less relevant measure for policy decisions around climate change.

How can we estimate climate sensitivity?

Climate sensitivity cannot be directly measured in the real world. Instead it must be estimated and there are three main lines of evidence that can be used to estimate it:

1. Historical climate records: instrumental records of warming since the mid-19th century, combined with estimates of greenhouse gas and aerosol emissions, can be used to assess the global temperature response to emissions of CO_2 by human activities to date.

2. Climate models: We can use climate models, which provide complex simulations of the Earth's climate system, to predict future climate sensitivity as we don't have observations for the future climate. These mathematical models are built around our understanding of the physics which underpin our climate system.

3. Palaeoclimate records: Ice cores and other records can be used to estimate natural changes in temperature and atmospheric CO_2 over thousands of years. These can be used for estimates of the past relationship between the two factors.

What is the current best estimate for climate sensitivity?

- The 2013 Intergovernmental Panel on Climate Change (IPCC) fifth assessment report suggests that ECS has a ‘likely’ range of 1.5 - 4.5°C. It adds ECS is extremely unlikely to be below 1°C and very unlikely greater than 6°C. The wide range accounts for all evidence.

- The IPCC estimate TCR has a likely range of 1.0 - 2.5°C and is extremely unlikely more than 3°C.

Role of Atmospheric Aerosols in Global Climate Change

What is aerosols? What is its source and composition?

- Aerosols in the troposphere are suspensions of liquid, solid, or mixed particles with highly variable chemical composition and size distribution. It can affect Earth's climate by scattering and absorbing the incoming solar radiation directly as well as by altering cloud amounts and properties. Aerosols, often called particulate matter (PM), are one of the major constituents of air pollution.
- Their variability is due to the numerous sources and varying formation mechanisms.
- Aerosol particles are either emitted directly to the atmosphere (primary aerosols) or produced in the atmosphere from precursor gases (secondary aerosols).
- Primary aerosols consist of both inorganic and organic components.
- Inorganic primary aerosols are relatively large (often larger than 1 μm) and originate from sea spray, mineral dust, and volcanoes.

Aerosols...........Contd.

- These coarse aerosols have short atmospheric lifetimes, typically only a few days.
- Combustion processes, biomass burning, and plant/microbial materials are sources of carbonaceous aerosols, including both organic carbon (OC) and solid black carbon (BC).
- BC is the main anthropogenic light-absorbing constituent present in aerosols.
- Its main sources are the combustion of fossil fuels (such as gasoline, oil, and coal), wood, and other biomass. Primary BC and OC containing aerosols are generally smaller than 1 µm.
- Secondary aerosol particles are produced in the atmosphere from precursor gases by condensation of vapours on pre-existing particles or by nucleation of new particles.

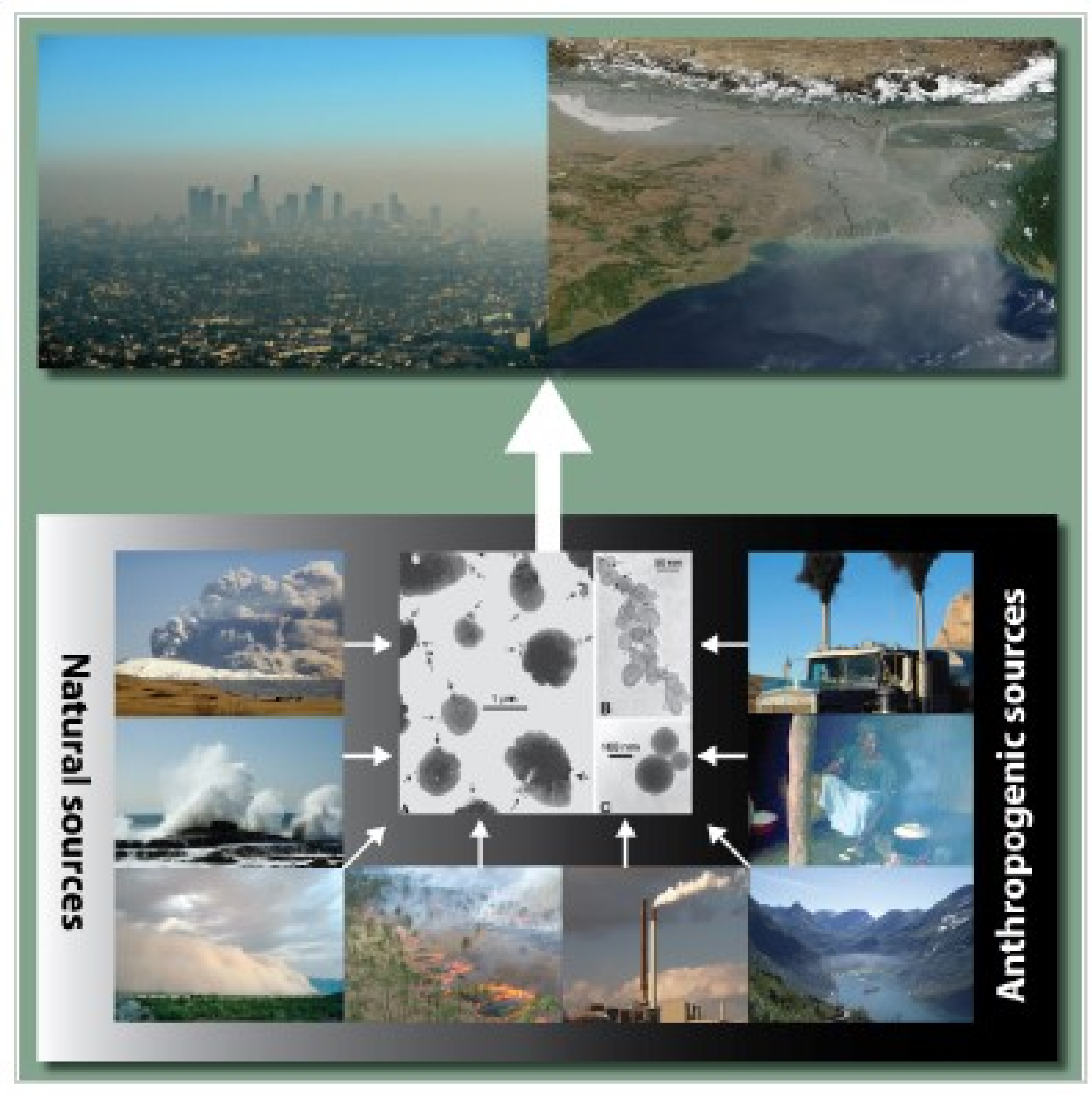
Aerosols...........Contd.

- A considerable fraction of the mass of secondary aerosols is formed through cloud processing.
- Secondary aerosols are small; they range in size from a few nanometres up to 1 μm and have long lifetimes.
- Secondary aerosols consist of mixtures of compounds; the main components are sulphate, nitrate, and OC.
- The main precursor gases are emitted from fossil fuel combustion, but fires and biogenic emissions of volatile organic compounds (VOCs) are also important. Occasionally volcanic eruptions result in huge amounts of primary and secondary aerosols both at the ground and in the stratosphere.
- Aerosol particles are so small that gravity has little effect on them.

Sources and Appearance of Atmospheric Aerosols

Sources include:

1. Volcanic eruptions (producing volcanic ash and sulphate),
2. Sea spray (sea salt and sulphate aerosols),
3. Desert storms (mineral dust),
4. Biomass burning (BC and OC),
5. Coal power plants (fossil fuel BC and OC, sulphate, nitrate),
6. Ships (BC, OC, sulphates, nitrate),
7. Cooking* (domestic BC and OC),
8. Road transport (sulphate, BC, VOCs yielding OC).



Aerosol (<5 μm radius) production sources and typical concentrations (μg m^{-3}).

A. Natural Sources	Production
1. Primary Production	
Sea Salt	2300
Mineral Particles	900-1500
Volcanics	20
Forest Fires & Biological Debris	50
2. Secondary production (gas particle):	
Sulphur from H_2S	70
Nitrates from NO_x	22
Converted plant hydrocarbons	25
Total Natural	~3687

Aerosol (<5 μm radius) production sources and typical concentrations (μg m^{-3}).....contd.

B. Anthropogenic	Production
1. Primary Production	
Mineral Particles	0-600
Industrial Dust	50
Combustion (Black Carbon) Organic Carbon	10 50
2. Secondary Production (Gas Particle):	
Sulphur from SO_2	140
Nitrates from NO_x	30
Biomass combustion (Organic)	20
Total Anthropogenic	~600

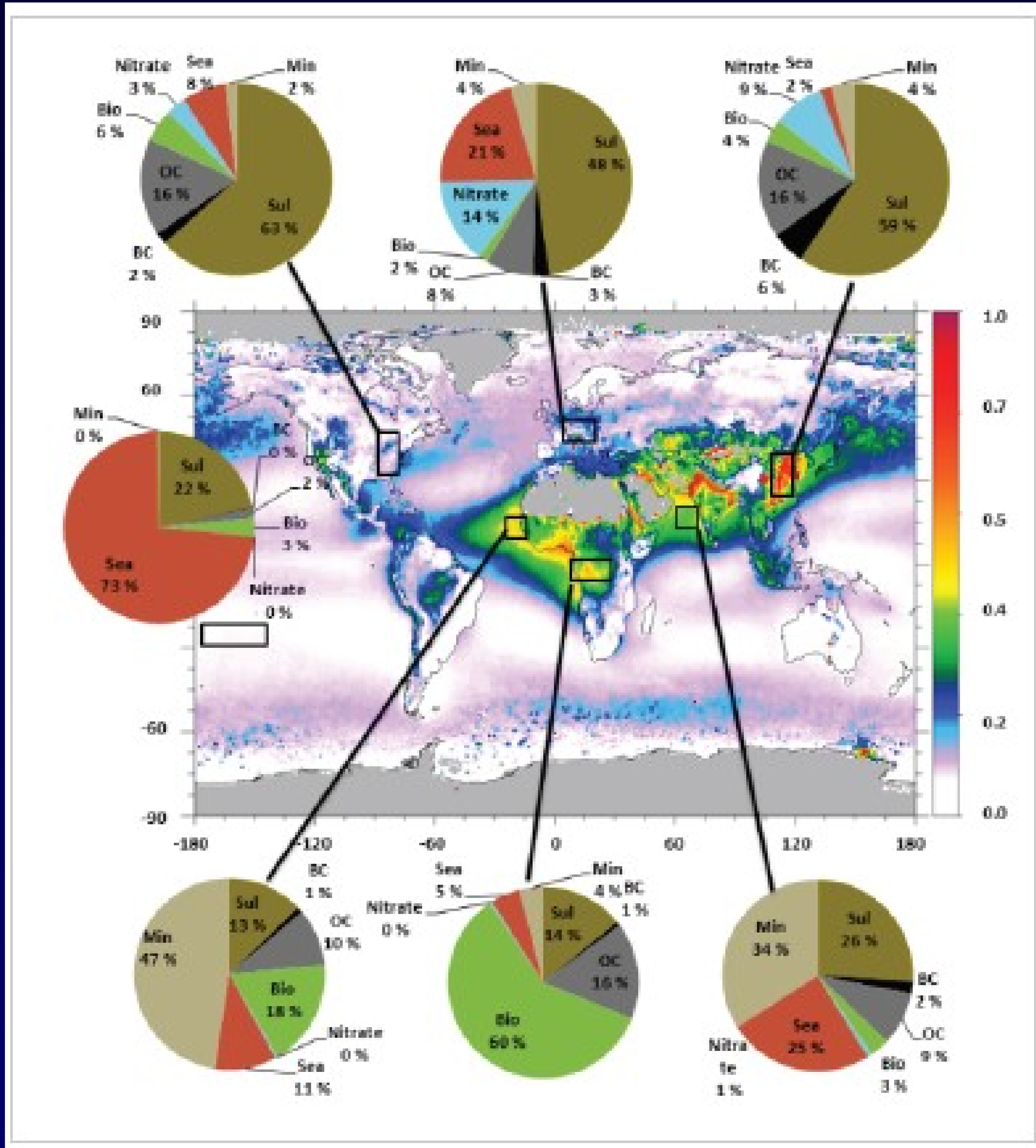
How are aerosols distributed globally?

- Aerosol optical depth (AOD) retrieved by remote sensing from space is highly inhomogeneous, with the largest values in Asia and the tropical regions of Africa.
- The estimated contributions from different aerosol types in selected regions reveal that in general, there is large spatial and temporal variability in global aerosol composition.
- Aerosol types are Sul (sulphate), BC and OC from fossil fuel usage, Bio (OC and BC from biomass burning), Nitrate, Sea (sea salt), and Min (mineral dust). Gray areas indicate lack of MODIS (Moderate-resolution Imaging Spectroradiometer) data.

Aerosol Optical Depth (AOD)

- AOD is a measure of the extinction of the solar beam by dust and haze. In other words, particles in the atmosphere (dust, smoke, pollution) can block sunlight by absorbing or by scattering light. AOD tells us how much direct sunlight is prevented from reaching the ground by these aerosol particles.
- It is a dimensionless number that is related to the amount of aerosol in the vertical column of atmosphere over the observation location.
- A value of 0.01 corresponds to an extremely clean atmosphere, and a value of 0.4 would correspond to a very hazy condition. An average aerosol optical depth for the U.S. is 0.1 to 0.15 and for India is ~0.40 to ~0.35.

Aerosols Distribution

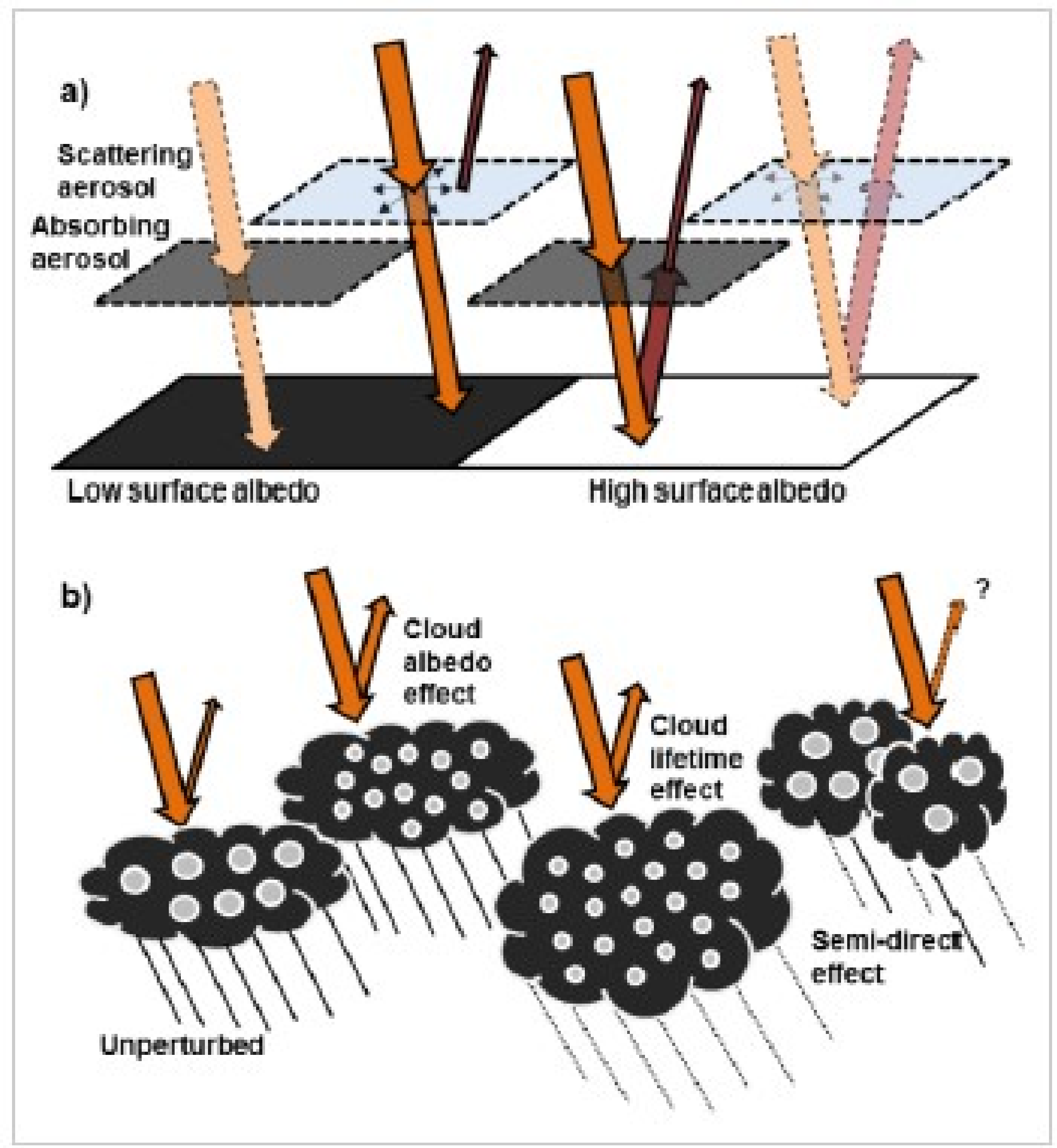


How do aerosols affect the climate?

- All atmospheric aerosols scatter incoming solar radiation, and a few aerosol types can also absorb solar radiation.
- BC is the most important of the latter, but mineral dust and some OC components are also sunlight absorbers.
- Aerosols that mainly scatter solar radiation have a cooling effect, by enhancing the total reflected solar radiation from the Earth.
- Strongly absorbing aerosols have a warming effect. In the atmosphere, there is a mixture of scattering and absorbing aerosols, and their net effect on Earth's energy budget is dependent on surface and cloud characteristics.
- Scattering aerosols above a dark surface and absorbing aerosols above a bright surface are most efficient. Scattering (absorbing) aerosol above a bright (dark) surface are less efficient because the solar radiation is reflected (absorbed) anyway.
- Absorbing aerosols are particularly efficient when positioned above clouds, which are a main contributor to the total reflection of solar radiation back to space.

The direct aerosol effect and the cloud albedo effect

(a) The direct aerosols effect for low and high surface albedo, for scattering and absorbing aerosols. A dark surface (low albedo) will already absorb a large portion of the solar radiation, and absorbing aerosols will thus have a small effect. Scattering aerosols will instead amplify the total reflectance of solar radiation, since the solar radiation would otherwise be absorbed at the surface. Over a bright surface (high albedo) scattering aerosols have a reduced effect. Absorbing aerosols may, however, substantially reduce the outgoing radiation and thus have a warming effect. (b) The cloud albedo effect (first indirect aerosol effect), cloud lifetime effect (second indirect aerosol effect), and semi-direct effect.



Role of Volcanism in Global Climate Change

Volcanism and Climate

- Our planet's climate results from a complex processes and events. The basic source of energy is radiation from the Sun. The incoming radiation interacts with the Earth's atmosphere and surface, so that changes to either can affect the climate.
- Volcanic eruptions are an important natural cause of climate change because tracer constituents of volcanic origin impact the atmospheric chemical composition and optical properties.
- But a much larger influence on climate comes from volcanic gases erupted into the atmosphere that spread out and encircle the planet.
- Major eruptions alter the Earth's radiative balance because volcanic aerosol clouds absorb terrestrial radiation, and scatter a significant amount of the incoming solar radiation, an effect known as "radiative forcing" that can last from two to three years following a volcanic eruption.
- Further, a dark lava flow from volcanic activity absorbs more of the solar energy than a desert soil, so a large enough lava flow could warm a local region.

Volcanism and Climate ....Contd.

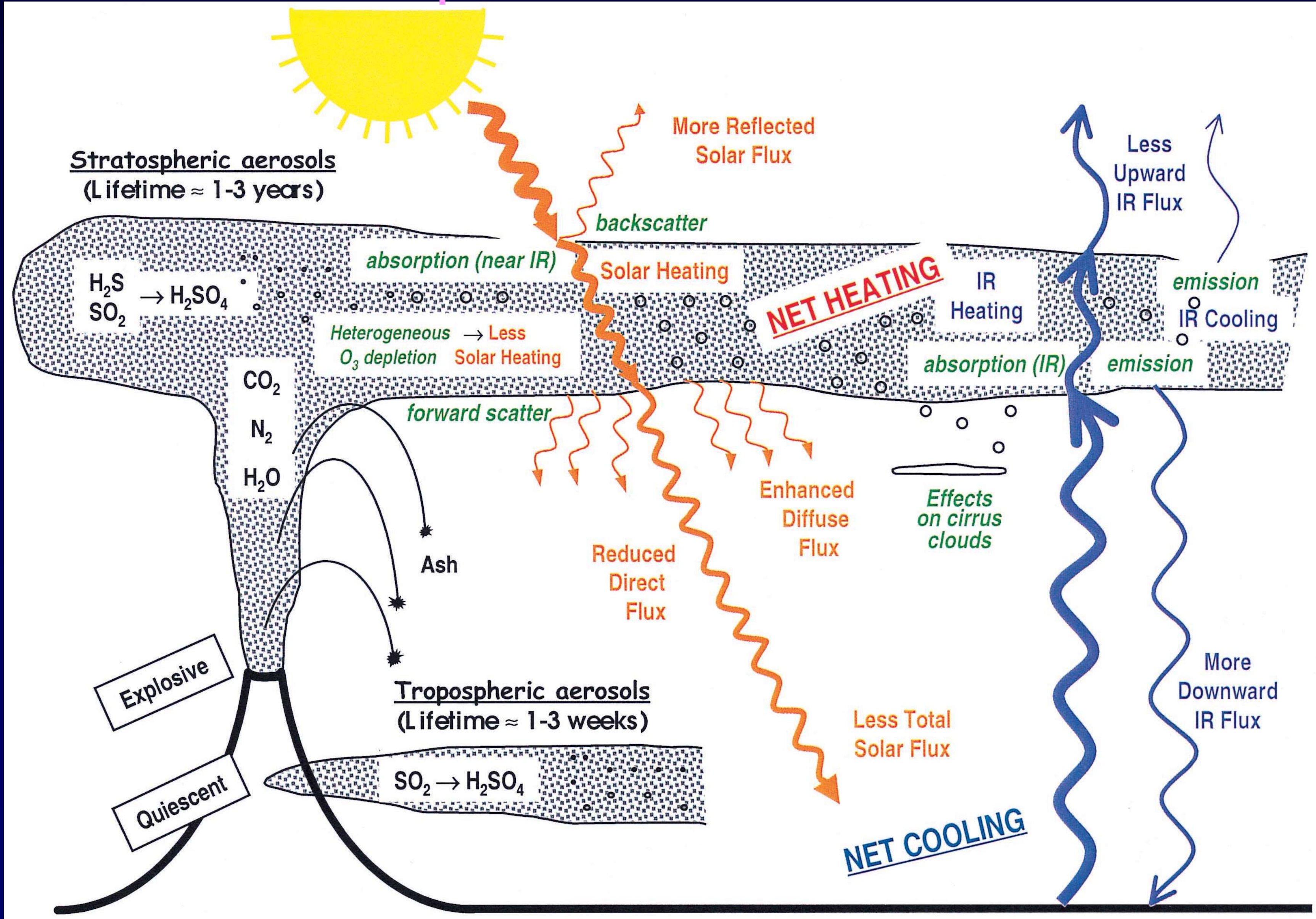
Volcanic Eruption

- The most abundant gas typically erupted is water vapor, which has been measured to be as high as 97% of gases erupted from some volcanoes.
- The water has very little impact on climate because it usually rains out of the atmosphere fairly quickly. In fact, it is very common to find volcanic ash deposited that preserve rainfall splash marks.
- The greenhouse gas carbon dioxide(CO_2) is the second most common gas(varying from 1% to 50% in different types of eruptions). Carbon dioxide is heavier than air and commonly ponds in low-lying areas. The volcanic CO2 does not significantly influence the climate because it is only about 1% of what is released by burning of fossil fuels.

Volcanism and Climate ....Contd.

- The gas that does have a noticeable climate impact is sulfur dioxide (SO_2). Unlike greenhouse gases, SO_2 cools the atmosphere.
- Magma contains a small amount of SO_2, typically less than 10% by volume. Large eruptions thrust the SO_2 into the upper atmosphere (the stratosphere) where it is transported around the planet.
- Contact with abundant water changes the SO_2 gas into sulfuric acid (H_2SO_4) droplets called aerosols. Even though they are microscopic, there are billions of such aerosols following a big eruption, so that they actually affect the climate.
- Each aerosol absorbs some of the radiation from the Sun, and thus heats itself and the surrounding stratosphere. But each ray of Sunlight that hits an aerosol does not strike the Earth, robbing the surface of that small amount of heat.
- During the 1900s there were three large eruptions that caused the entire planet to cool down by as much as 1°C. Volcanic coolings persist for only 2 to3 years because the aerosols ultimately fall out of the stratosphere and enter the lower atmosphere where rain and wind quickly disperse them.

Schematic diagram of volcanic inputs to the atmosphere and their effects



Volcanism and Climate ....Contd.

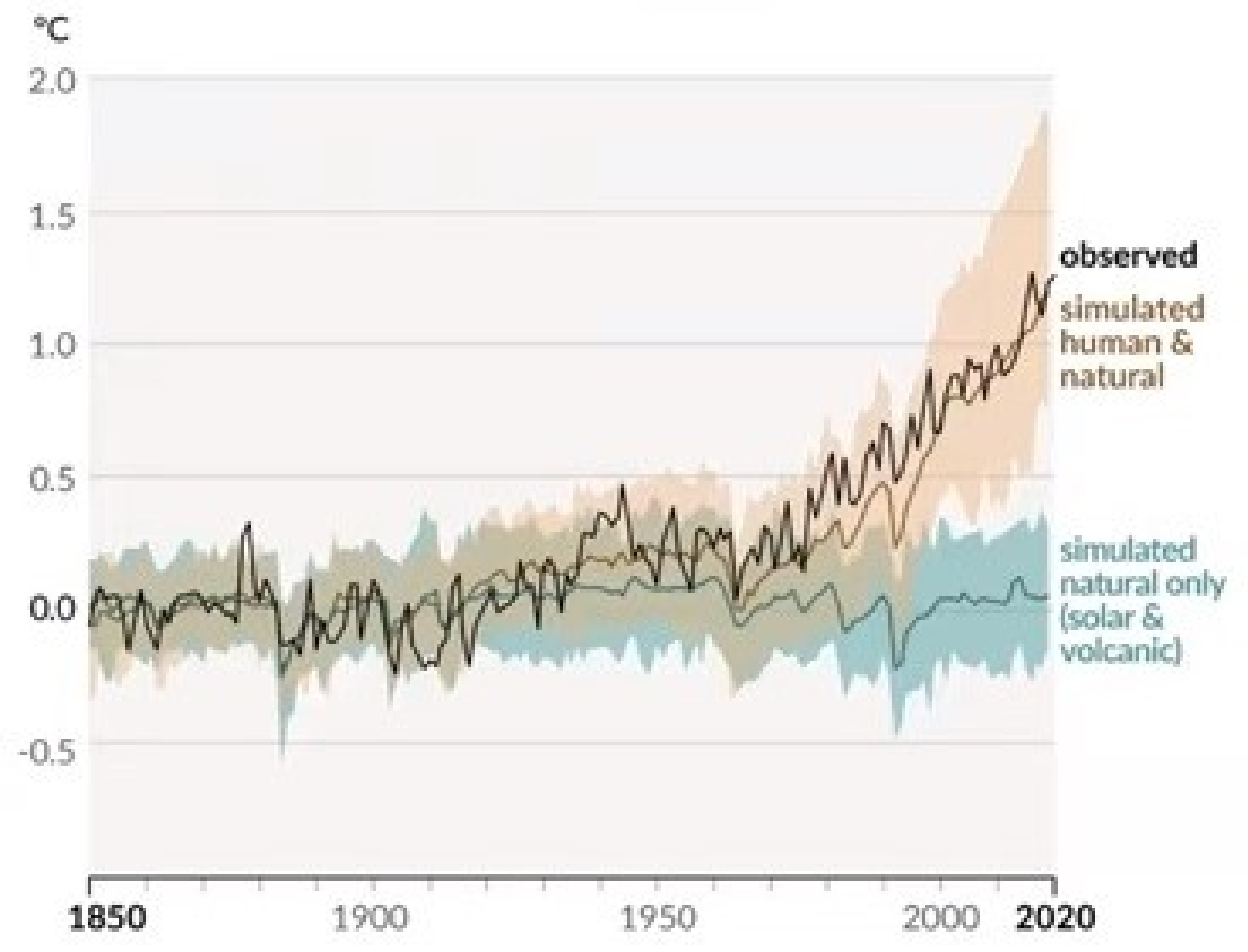
Although researchers understand the basic mechanism of cooling due to eruptions there are many details still to be investigated. Here are some questions that need to be investigated:

- Not all volcanic eruptions seem to effect climate. What are the characteristics, other than bigness, of those that effect climate change?

- How can one predict which volcanoes are likely to impact the climate when they erupt?

- What would be the climate effect if a series of large eruptions occurred over 10 years?

Change in global surface temperature relative to 1850-1900

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